

Financial Zoology and Hedge Funds

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Outline

- 1 Zoology of Financial Markets
- 2 Hedge Funds
- 3 Leverage, short-selling and derivatives
- 4 Conclusions

1 Zoology of Financial Markets

2 Hedge Funds

3 Leverage, short-selling and derivatives

4 Conclusions

Financial markets as a biological system

Why a “Zoology” of capital markets?

- All financial agents have their own particular niche in the market: **species**.
- All markets have infrastructure (geography), regulations (Nature constraints), investment opportunities (food) and agents (species): **ecosystem**.
- All agents are related, either in a cooperative or a competitive way: **symbiosis, parasitism and food chain**.
- Agents can change their strategies to enhance profit, and only the best ones remain: **mutation and natural selection**.
- When market rules change or new players come in, all agents need to adapt or die: **invasions, evolutionary adaptation and extinction**.

Description of financial agents I

Retail banks

- They **pay** the depositors' savings at a **lower interest rate**.
 - They **loan** at a **higher interest rate**.
 - Their profit is the difference between both interest rates.
 - The idea is to make capital flow where it is needed: from savers to entrepreneurs.
- ⇒ Risk management = **creditworthiness**.
- Business model is not complex nor sophisticated: **dull banking**

Investment banks

- They sell financial instruments to investors: **selling side**.
- They create new products under investors' demand.
- Complex products require strong risk management policies: **quants**.
- Profit made by volume and margin/prime over the hedging price of the product: **flow trading**.

Description of financial agents II

Markets

- A market is where a buyer of an instrument meets a seller.
- It can be an exchange or an over-the-counter (OTC) market.
- A transaction in an **exchange**:
 - ★ Is done with **listed or standard instruments**, e.g. futures or listed options.
 - ★ Has a **clearing house**: clearing margins and daily settlements.
 - ⇒ No counterparty or credit risk.
- A transaction in an **OTC market**:
 - ★ Is done with **any imaginable instrument** e.g. forwards or exotic options.
 - ★ But has **no guarantee** that the trading parts will honour the agreement.
 - ⇒ Counterparty or credit risk.

Description of financial agents III

Brokers

- They are **middlemen**: link between investors and financial markets.
- In general they participate in exchanges, but they can also have OTC clients.
- They charge a **fee**: fixed (account management) and/or per contract traded (unitary price or percentage of whole transaction).
- Their **good reputation** is a guarantee of the trade: investors who do not know each other can buy or sell assets in full confidence via the broker.
- Profit made **by volume**.

Description of financial agents IV

Market-makers

- They are agents who buy and sell assets in financial market.
 - They make **firm quotes**: once they show a price and a volume, they are bound to trade at those conditions if there is a willing counterparty.
- ⇒ **Liquidity providers**.
- Their buying (bid) price is smaller than their selling (ask) price.
- ⇒ On each buy-and-sell transaction **they earn the spread**: the difference between the ask and the bid prices.

Small savers

- They **put their money in retail banks**, earning an interest rate.
- If there is **trust in the bank**, savers leave their money inside; if there is no trust they withdraw their money.
- When several savers withdraw their money at the same time, the bank can become insolvent or bankrupt: **bank run**.

Description of financial agents V

Small investors

- People with **less than 100k USD**.
- They do not have access to all financial services directly: they go through middlemen (e.g. brokers and investment banks).
- They have an information disadvantage with respect to financial services providers $\Rightarrow$ **more legal protection**.

Qualified investors

- Legal persons (individuals) with **more than 250k USD**.
- They are considered to have the same information and sophistication than any financial services provider $\Rightarrow$ **less legal protection**.
- They can access complex investment strategies with **higher return** (but potentially higher risk).
- They tend to search for high returns with a short- and middle-term investment horizon (less than 5 years).

Description of financial agents VI

Institutional investors

- They are legal entities (not individuals) with **capital over 100M USD**.
- They are as **sophisticated as qualified investors**: access to all asset classes and investment strategies.
- They have **restrictions and controls** due to law and internal mandates: pension funds, sovereign funds, insurers, reinsurers, etc.
- For (re)insurers, this will change with **Solvency II**: they will be more like asset management.
- They tend to diversify their risk and look for stable long-term returns (10+ years).

Description of financial agents VII

Asset management

- They are **professionals of portfolio management** (assets or baskets of assets).
- They invest on financial instruments **on behalf of their clients**: qualified investors, institutional investors.
- In some cases, they can manage money from small investors via **pooling, investment vehicles**, etc. . .
- They charge a **fee** on the amount of money invested in them or AUM (**Assets Under Management**).
- Depending on their **mandate** they can use different strategies with different investment horizons.
- They have **risk management** policies and can use different **risk measures**: Sharpe Ratio, Maximum Drawdown, Value-at-Risk (VaR), Expected Shortfall (CVaR), stress tests, Monte Carlo, etc.

Description of financial agents VIII

Prop traders

- They speculate in capital markets with **their own money**.
- ⇒ They have the **highest flexibility on investment strategies**: derivatives (vanilla, exotic, structured), high-frequency trading, etc. . .
- Before the **Dodd-Frank reform**, in particular the **Volcker Rule**:
 - ★ Big commercial banks used to have their own prop trading desks: dedicated teams speculating on behalf of the firm.
 - ★ Prop trading teams were legally and accountably separated from the rest of the investment bank.
 - ★ Now prop trading in banks is almost banned, but **market-makers** in banks are exonerated of the Volker rule.

Description of financial agents IX

Arbitrageurs

- An **arbitrage** is to take advantage of a market inefficiency.
- An **arbitrageur** is generally a speculator, but their role is to make markets more efficient.

Examples of arbitrage:

- Prices or **mispricing arbitrage**: buy and sell instruments that are far from their *fundamental price*.
- Venues or **space arbitrage**: buy and sell the same product in two different markets, cashing the price difference.
- Time value or **term-structure arbitrage**: interest rates and volatility, calendar strategies with options.
- Statistical arbitrage or **stat arb**: spot and exploit price patterns that are statistically robust.

Description of financial agents X

Prime brokers

Prime brokers or **credit lines** are the bankers of investment management firms. The services they offer are:

- **Borrow money** to fund managers, so they can buy assets.
- **Borrow assets** to fund managers, so they can **short-sell** them.
- They assume the counterparty risk of the fund: their reputation and **due diligence** are vital.
- They offer legal, accountable and commercial **advisory** and support.

Description of financial agents XI

Hedge funds

- They use **sophisticated and innovative** investment strategies:
 - ★ Technical and financial knowledge above competitors.
 - ★ More rigorous risk management than traditional asset managers.
 - They are very active i.e. high **turnover**, unlike the classic *buy-and-hold* strategies.
 - They charge high fees, both on AUM and on performance, but they guarantee an **absolute return**.
- ⇒ Very **aggressive business model**: only 50% survive after 3 years of launch.
- Generally they are small and look for investment niches: **small is beautiful**.

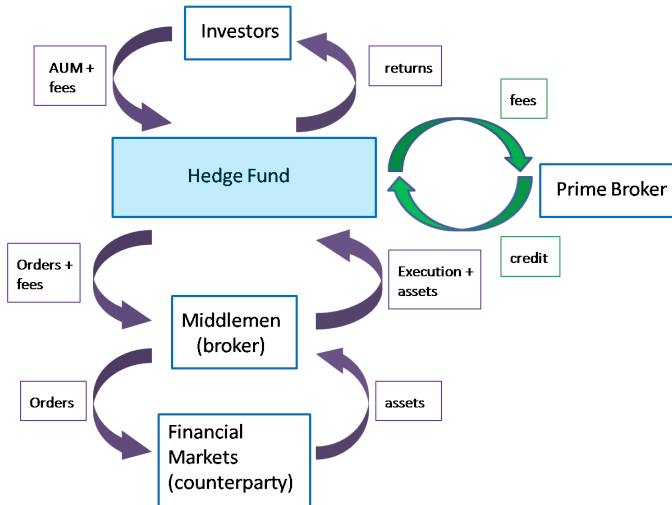
In summary ...

Zoology of the Financial Markets

- Retail banks.
- Investment banks.
- Markets: exchanges and OTC.
- Middlemen: brokers and market-makers.
- Small savers (retail agents).
- Small investors.
- Institutional investors e.g. pension and sovereign funds.
- Portfolio / asset managers.
- Prop traders.
- Arbitrageurs.
- Prime brokers.
- Hedge funds.

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Hedge funds and their ecosystem



Different return philosophy I

Capital Asset Pricing Model (CAPM): alpha and beta

The **alpha and beta** of a financial asset (or investment fund) are defined as

$$r_A = \alpha + \beta r_M + \varepsilon,$$

where r_A is the return of the asset (or fund), r_M the market return and ε is a zero-mean error (orthogonal to r_M).

- βr_M is the part of the return that is explained by the market return r_M and the correlation between r_A and r_M : **β is the market exposure**.
- α is the part of the return that is not explained by market fluctuations: **the Holy Grial** of the asset management.
- According to the Efficient Market Theory (EMT), we should have $\alpha = 0$.

Different return philosophy II

Relative return

It is the **extra return with respect to a benchmark**, e.g. the market return r_M .

- β is a measure of market exposure: if $\beta > 1$ the asset is called **aggressive or cyclical**, if $\beta < 1$ it is **defensive or non-cyclical**.
- α measures how skilled is the manager: it is the excess return over the market exposure.
- In the traditional asset management, as long as $r_A > r_M$ the manager is doing a good job: **stock-picking**.
- However, one could have that $0 > r_A > r_M$.
- But the manager is happy nevertheless: -10% is a relative return of $+5\%$ with respect to a -15% benchmark, right?
- But **what is in reality the market return**? Indices, trackers, ETFs, etc.

Different return philosophy III

Absolute return

It is a **return without benchmark**: the mantra of all hedge funds.

- The goal is to deliver **positive returns, uncorrelated with the market**.
- ⇒ Protect or **hedge** the money of investors.
- In other words, the absolute return aims for **$\beta = 0$ whilst maximising α** .
- But in order to have a zero market exposure, the manager needs a minimum amount of liberty to enter and exit her positions.
- ⇒ Aggressive, innovative investment strategies with rigorous risk management.

Fees I

The 2% rule

- A traditional investment fund charges an **annual fee of 2% on the total value of Assets Under Management (AUM)**.
- For example, if an investor buys 100M EUR of the fund, the manager takes a 2M EUR cut for managing the account.
- If the fund's return is at least 2% above the benchmark, the fee is justified:
 - ★ Otherwise, the investor has a negative net return vs the benchmark.
- For competitiveness, the fund can apply a smaller AUM fee.

Fees II

The 2/20 rule

- A hedge fund also charges a fee of 2% of the AUM.
- But there is a second fee that distinguishes hedge funds from traditional asset managers: the **performance fee**.
- The performance fee can vary between 10% and 20%, but **20% is standard** in the industry.
- For example, suppose a hedge fund has a **total return of 10%**:
 - ★ The **client** gets a net return of $8\% = 10\% \times (1 - 0.2)$.
 - ★ The **hedge fund** pockets $2\% = 10\% \times 0.2$.
 - ★ Total fee collected by the hedge fund is $4\% = 2\% \text{ AUM} + 2\% \text{ performance}$.

⇒ Twice the fee of a traditional asset manager.

Fees III

The *high-water mark* and *hurdle rate* rules

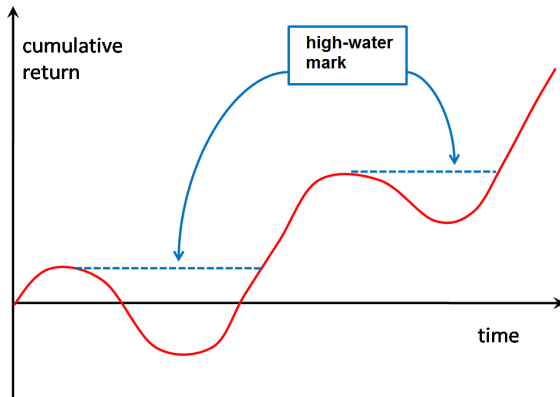
Since the 2/20 fee structure is high, investors demand guarantees or insurances:

- The 20% performance fee only triggers if the cumulative return of the fund is above the latest peak: *high-water mark*.
- The performance fee triggers only if the return is above a certain threshold or *hurdle rate* e.g. LIBOR (London Interbank Offered Rate) or T-notes (US Treasury).

The business model of hedge funds is very aggressive

- They need to deliver sustainable returns above both the hurdle rate (e.g. market) and other asset managers.
 - An unlucky year + high-water mark can block the performance fee for years.
- ⇒ Low survival rate of hedge funds.

Fees IV



Convergence of interests

Hedge funds: interests aligned with investors'

- The investors leave their money in a common account: **pooling**.
 - In a hedge fund, managers and founders put their money in the same pooling: **skin in the game**.
 - The bonuses of employees are blocked for several years, but each year the corresponding amount is deposited in the pooling.
 - Moreover, the performance fee triggers only if:
 - ★ The return is above the hurdle rate.
 - ★ The cumulative return is above the high-water mark.
- ⇒ Both investors and hedge fund managers are on the same boat.

Light-handed regulation

The 3 wishes of a hedge fund

- **Light red tape:** to access different capital markets and reach more potential investors.
- **Low taxes:** the lower the taxes, the bigger the net returns for the investors (and the fund).
- **Liberty of management:** to access all possible investment strategies to maximise α and minimise β .

In the search of the promised land

Hedge funds are accountably located in tax-friendly locations:

- **Off-shore.** Tax havens outside the jurisdiction of US and EU: e.g. Bahamas, Bermuda, Cayman Islands, Virgin Islands.
- **On-shore.** Low-tax zones within US and EU: Delaware, Ireland, Luxembourg, Switzerland, Cyprus.

But managers are physically located in financial centres: NY, London, Paris, etc.

Limited access

For qualified investors only

To keep their legal exemptions and as a measure to protect small investors, hedge funds need to satisfy 2 conditions:

- **Zero marketing**: it is the investors who solicit the hedge fund, not the other way around.
 - ★ This is about to change: the SEC lifted the marketing ban on July 2013.
- **Only for high-net-worth individuals**:
 - ★ at least 1M EUR of assets,
 - ★ or an annual salary above 200k USD in the last 2 years (300k USD for married couples),
 - ★ and with a minimal investment in the hedge fund of 125k EUR.

The goal is that only those individuals with sufficient financial knowledge and money should invest in hedge funds.

Aggressive and innovative investment strategies I

Hedge funds are heterogeneous

According to Hedge Fund Research Inc., hedge funds can be:

- **Equity:** market-neutral, fundamental (growth/value), quantitative directional, sector-based, short bias.
- **Event driven:** activist, credit arbitrage, distressed/restructuring, merger arbitrage, private equity.
- **Macro discretionary or systematic:** active trading (HFT), commodities, currency.
- **Relative value:** asset-backed, convertible arbitrage, fixed-income (corporate or sovereign), volatility, real estate.
- **Multi-strategy:** a mix of several of the previous strategies.
- **Fund of funds:** a portfolio consisting on several hedge funds.

Aggressive and innovative investment strategies II

The 3 jewels of a hedge fund

- **Leverage:**

- ★ It allows to take positions worth more than the AUM.
- ★ It **amplifies profits**: $\text{leverage} = \ell \Rightarrow \text{profit} = \text{return} \times \ell$.
- ★ But it also **amplifies losses**: handle with care.

- **Short-selling:**

- ★ It is commonly referred as “selling what you do not own”, but is more complex than that.
- ★ The fund can only sell those assets the prime broker borrows, paying a fee (interest rate).
- ★ It allows to buy and sell assets: **ride the wave** upward or downward.

- **Derivatives:**

- ★ They provide profits under specific market conditions: **pay-off**.
- ★ They have **leverage** stronger than normal assets.
- ★ But beware: one can lose more than the initial bet e.g. naked options.

In summary, what is a hedge fund?

A hedge fund is an investment fund . . .

- 1 With a different philosophy of return: absolute return.
- 2 With performance fees, high watermark and hurdle rate.
- 3 With managers and investors sharing the same goal: skin in the game.
- 4 With light regulation: located in low-tax zones.
- 5 With limited liquidity and access: only qualified investors and withdrawals at specific dates.
- 6 With diverse investment strategies but always using leverage, short-selling and/or derivatives.

Remarks on the hedge fund industry I

In theory, hedge funds are not *market beaters*

- **Absolute return** means positive returns in ALL market conditions.
- **Hedge** means to protect/cover positions, reducing market exposure.
- **Alpha** means returns decorrelated from market fluctuations.
- By **market efficiency** $\Rightarrow$ hedge funds are expected to have **reduced returns** when markets are bullish.

In practice, hedge funds can be *market beaters*

Using alpha, beta and leverage, **hedge funds can beat the market** systematically:

- In bullish markets $\Rightarrow$ maximise and leverage beta.
 - In bearish markets $\Rightarrow$ minimise beta whilst maximising and leveraging alpha.
 - If alpha is big enough and beta is small $\Rightarrow$ leverage alpha and forget about beta.
- $\Rightarrow$ **No hedging** but rather making **leveraged, speculative bets**.
- $\Rightarrow$ An investment too good to be true is probably so, e.g. bubbles, subprimes, Madoff.

Remarks on the hedge fund industry II

Performance 2013 for Standard & Poors (S&P)

- S&P500 is the index of the 500 biggest market capitalisations with common stock listed on NYSE and NASDAQ.
 - S&P500 made 29.60% in 2013, or 32.39% including dividends.
 - An S&P500 tracker (traditional fund) ask fees around 0.70% for retail investors and 0.35% for institutional investors.
 - An S&P500 ETF (exchange-traded fund) ask fees around 0.40% for everyone.
- ⇒ Investing in **trackers/ETFs** in 2013 would have yielded **29%**.

Hedge funds have not been very successful lately

- The last time hedge funds beat the market was in 2008: -19% vs -37%.
- In 2013 hedge funds made 7.4% of net average return.
- Only 10 out of the 100 best large (+1bn USD) hedge funds outperformed the S&P500.
- Although the best hedge fund made 84.2%.

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Leverage I

Definition

- The **leverage** is the quotient of the total value of the fund's positions over the total value of the fund's assets:

$$\text{leverage} = \frac{\text{positions}}{\text{AUM}}$$

- For example, if a hedge fund has 1M EUR of AUM and a total exposure of 3M EUR then its leverage is 3.
- During the 2007 crisis, the investment banks had a leverage of 30.
- The **Basel II** rules require banks to hold **liquid assets** of at least 8% of their total positions, i.e. **a leverage of maximum 12.5**.
- However, **Basel III** requires at least 10.5% of **liquid assets**, i.e. **a leverage of maximum 9.5**.
- Historically, hedge funds have been below 5 (currently it is around 1.5).
- But **what is a liquid asset** anyway?

Leverage II

Example 1: buying at margin

- Suppose the hedge fund has 1M EUR of AUM.
- The prime broker authorises a credit line of 2M EUR, i.e. a **50% margin or leverage = 2**.
- The fund buys 2M EUR of a stock of unitary price of 10 EUR, i.e. 200k stocks.
- What happens if the **the stock goes up to 10.10**?
 - ★ Profit of 10% in the market.
 - ⇒ Profit of $10\% \times 2 = 20\%$ for the fund.
- But what if **the stock goes down to 9.90**?
 - ★ Loss of 10% in the market.
 - ⇒ Loss of $10\% \times 2 = 20\%$ for the fund.
- If the prime broker authorises a **margin of 5% i.e. a leverage = 20**:
 - ★ A profit of 10% in the market ⇒ a profit of 200% for the fund.
 - ★ But a loss of 10% in the market ⇒ a loss of 200% for the fund.

Leverage III

Example 2: carry trade

- The hedge fund opens a credit line with its prime broker to:
 - ★ Buy 1M USD of high-yield bonds e.g. Mexico's at 4%.
 - ★ Sell 1M USD of low-yield bonds e.g. US at 1%.
- The hedge fund pockets the difference or **spread**

$$4\% - 1\% = 3\%$$

- If the fund has a leverage of 5 (i.e. a 20% margin) then its profit is:

$$3\% \times 5 = 15\%$$

Risks on carry trade:

- The high-yield bond has a higher credit risk than the low-yield bond: market prime of risk.
- If the spread is too small, small changes in monetary policies can kill the deal.

Leverage IV

Risks of leverage

Leverage is a double-edged sword since **one can lose more than the initial bet**:

- It is not for everyone, only knowledgeable investors should use it.
- But there are a lot of websites for retail trading with **leverage above 400**:
 - ⇒ For example, a French Forex broker offers a credit line of 800k EUR vs 2k EUR of deposit.
 - ⇒ A drop of 0.5% on the traded currency implies a loss of 200% for the trader.

When retail traders lose more money than what they bet because they do not understand what is leverage, who is to blame?

Moral of the tale

- A trader needs to understand the risks of investing in a leveraged instrument.
- Otherwise, they can get burnt as Icarus.

Short-selling I

Definition

- A **short sell** is to sell an asset now with the idea of buying it back later at a smaller price, making a profit.
 - But short-selling is more than just *selling what you do not own*:
 - ★ The fund has to borrow the asset to its prime broker.
 - ★ The fund pays a fee (interest rate) for borrowing the asset.
 - ★ The fund has to return the borrowed asset when the prime broker wants it back.
- ⇒ Short-selling is profitable for the fund only if the expected return is greater than the prime broker's interest rate.

Short-selling II

Example 1: short-selling stocks

- A hedge fund bets that Adidas will end the day at a loss.
- The hedge fund borrows from his prime broker 1M EUR worth of Adidas stocks.
- Assume that the fund pays a fee of 0.1% for borrowing the stock:
 - ★ If Adidas goes down 1% the fund gets a profit:

$$-(-1\%) - 0.1\% = +0.9\%$$

- ★ But if Adidas goes up by 1% the fund loses:

$$-(+1\%) - 0.1\% = -1.1\%$$

⇒ **Asymmetric bet:** With short-selling there is more risk than with long-only positions.

Short-selling III

Example 2: leverage and long/short

- A hedge fund has 1M EUR of AUM
- The fund opens a credit line with its prime broker to:
 - ★ Buy 2M EUR of L'Oréal: *long leg*
 - ★ Sell 1M EUR of Carrefour: *short leg*

⇒ The leverage is thus

$$\frac{2M + 1M}{1M} = \frac{3M}{1M} = 3$$

- A **long-only fund** cannot short-sell: the only way to beat the market is to pick stocks that outperform the market.
- **Traditional long-short funds** follow the 150/50 rule: up to 150% of AUM in long positions and up to 50% in short positions ⇒ leverage = 2.
- **Hedge funds** have total liberty on how much they can long or short, as long as they respect the leverage authorised by their prime broker.

Short-selling IV

Example 3: market-neutral portfolios and pair trading

- A **pair-trading** strategy consists in buying one asset and selling another, both strongly correlated: e.g. HSBC and Barclays.
- A **market-neutral portfolio** P is built by buying x shares of HSBC and selling y shares of Barclays, such that the market exposure of P is zero:

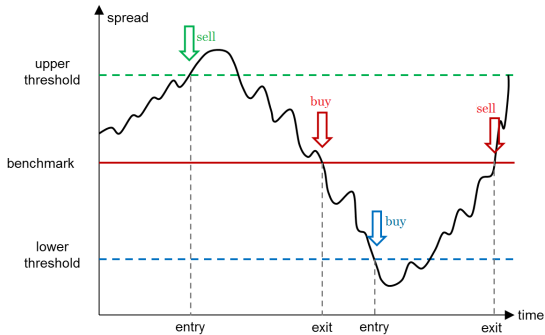
$$\beta_P = x\beta_{HSBC} - y\beta_{Barclays} = 0$$

- If HSBC and Barclays are strongly correlated then they tend to move together:

$$\beta_{HSBC} = \beta_{Barclays}$$

- The **spread** HSBC–Barclays consists on buying 1 share of HSBC (long leg) whilst selling 1 share of Barclays (short leg).
- The **bet** is that the spread has a **mean-reverting dynamic**: HSBC and Barclays can lose momentarily their correlation, but the market will eventually correct it.

Short-selling V



One of many *stat arb* strategies: mean-reverting pair-trading.

Derivatives I

Properties

- They have an intrinsic leverage effect: options, CDS, CDO, etc.
 - They allow the investor to have specific return profiles: **payoff**.
- ⇒ Huge potential of diversification and hedging.
- ⇒ They need a rigorous and sophisticated risk management: dynamic hedging to control the sensitivities to the market or **greeks**.
- They can be standardised (*listed*) or customised (*over-the-counter*).
 - They can be as complex as the investment bank that sells them and the investor who buys them can afford.
- ⇒ **Realistic and robust mathematical models** are crucial.

Derivatives II

Example 1: intrinsic leverage effect of derivatives (1/2)

Suppose we buy a call option on a stock with the following parameters:

- Spot price $S = 100$ EUR.
- Strike $K = 100$ EUR.
- Maturity T months.
- Two possible outcomes at time T :

$$\mathbb{P}[S(T) = 101] = \mathbb{P}[S(T) = 103] = 1/2$$

⇒ Price of the option is **2 EUR**:

$$\mathbb{E}[(S(T) - K)_+] = (101 - 100) \times 1/2 + (103 - 100) \times 1/2 = 2$$

Derivatives III

Example 1: intrinsic leverage effect of derivatives (2/2)

Suppose that $S(T) = 103$ EUR:

- If we bought the **asset** then the return is **3%**: +3 EUR from 100 EUR invested.
- If we bought the **option** then the return is **50%**: +1 EUR from 2 EUR invested.

But if $S(T) = 101$ EUR:

- The **asset** return is **1%**: +1 EUR from 100 EUR invested.
- The **option** return is **-50%**: -1 EUR from 2 EUR invested.

Derivatives IV

The “Greeks”

Every derivative product (e.g. an option) has several parameters:

$$\text{Derivative} = C(S, t, K, T, r, \sigma)$$

For each parameter, a sensibility or **greek** is computed:

- Spot price: $\frac{\partial C}{\partial S} = \Delta$ (delta)
- Time: $\frac{\partial C}{\partial t} = \Theta$ (theta)
- Delta: $\frac{\partial^2 C}{\partial S^2} = \Gamma$ (gamma)
- Interest rate: $\frac{\partial C}{\partial r} = \rho$ (rho)
- Volatility: $\frac{\partial C}{\partial \sigma} = \mathcal{V}$ (vega)

Derivatives V

Example 2: market-neutral portfolios and derivatives

- An investor buys/sells a derivative C : a vanilla/exotic option, a CDS, etc.
- The derivative's sensitivity to the market is measured by Δ , i.e. the fluctuations of the asset price S .
- In order to **hedge the derivative**, the investor buys a (signed) quantity Δ of shares.
- A portfolio is **market-neutral** if it is hedged against market risk.
- The total portfolio $P = C - \Delta S$ is market-neutral, i.e. insensitive to fluctuations in S :

$$\begin{aligned}\Delta(P) &= \partial P / \partial S \\ &= \partial C / \partial S - \Delta = 0.\end{aligned}$$

Derivatives VI

Imperfect Δ -hedge

The market-neutrality of a portfolio P via Δ -hedge is not perfect:

- **First-order hedge**: $\Delta(P)$ is zero but not necessarily $\Gamma(P)$.
 - The Δ -hedge has to be done **infinitely often**: impossible because of transaction costs (broker's fees, taxes, overnight interest rates, etc).
 - Δ can be any number, but in practice it has to be **an integer**: we cannot buy/sell 3.1416 shares, can we?
- ⇒ The Δ -hedge of a trader is done within the limits of a **risk budget**, i.e. the minimum and maximum exposure in currency (not in shares):

$$\Delta^{\min} < \Delta < \Delta^{\max}.$$

- But **how is the risk budget defined**? Ask your risk manager.

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Final comments

Summary of this presentation

- We described the financial zoology and how all different agents interact in this competitive, highly adaptive ecosystem.
- We explained what is a hedge fund and distilled their main characteristics.
- We saw in detail the investment tools of a hedge fund: leverage, short-selling and derivatives.

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